



TEST CODE **02120020**

MAY/JUNE 2004

FORM TP 2004197

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION
ENVIRONMENTAL SCIENCE
UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE
PAPER 02

2 hours 30 minutes

21 MAY 2004 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Answer **ANY TWO** questions from **EACH** Module; a total of **SIX** questions.
4. Write your answers in the answer booklet provided.
5. The use of a non-programmable calculator is allowed.

MODULE 1

Answer ANY TWO questions.

1. Two species of protozoa, A and B, were grown in a laboratory. First, they were grown separately and then together under the same conditions. Figure 1 below shows how the populations of the two species changed over time.

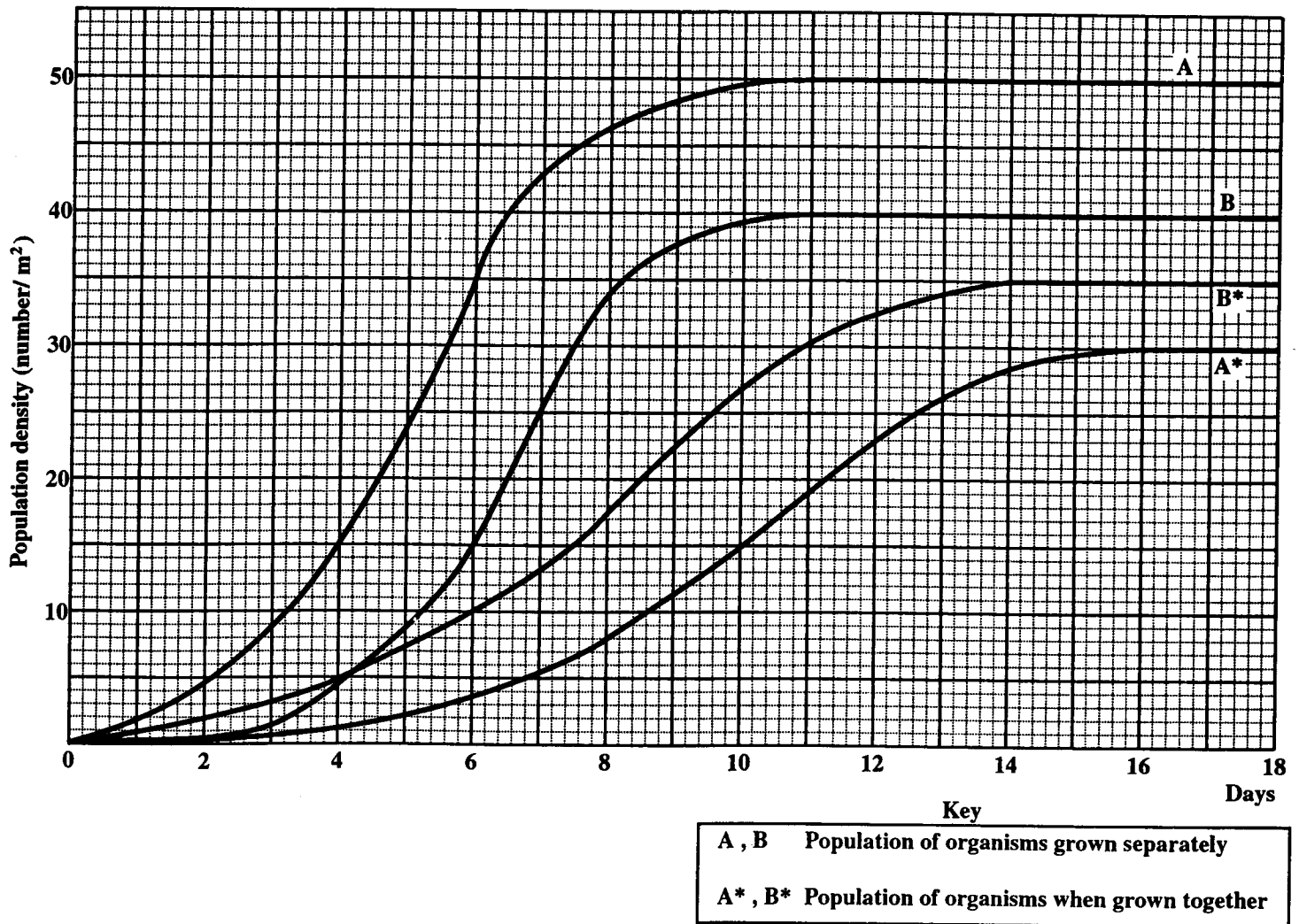


Figure 1. Population of two species of protozoa

- (a) Distinguish between 'habitat' and 'niche'. [4 marks]
- (b) With reference to Figure 1, state FOUR observations regarding the change in the population of BOTH species over time. [4 marks]
- (c) Outline FOUR possible reasons for the observations made in (b) above. [12 marks]

Total 20 marks

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2. (a) Outline FOUR differences between primary and secondary ecological succession. [8 marks]
- (b) Discuss the relationship between ecological succession and climax communities in the stability of ecosystems. [6 marks]
- (c) Describe TWO ways in which human activities have affected the nitrogen cycle. [6 marks]

Total 20 marks

3. (a) Give FOUR reasons why, in an ecosystem, a food web is generally more useful than a food chain. [4 marks]
- (b) Outline FOUR reasons for assigning trophic levels to each organism in an ecosystem. [8 marks]
- (c) With reference to Figure 2 below, explain why it is more energy efficient for humans to eat at lower trophic levels.

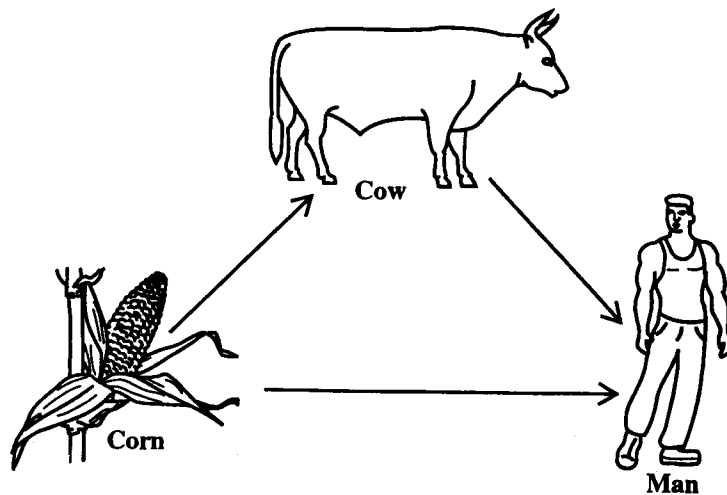


Figure 2. A food chain

[8 marks]

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. (a) Define 'total fertility rate'. [2 marks]
- (b) (i) Discuss FOUR factors that influence total fertility rate. [8 marks]
- (ii) How does EACH factor impact on population growth rate. [4 marks]
- (c) Assess how ONE of the factors in (b) (ii) above contributes to the population rate of growth in your country. [6 marks]

Total 20 marks

5. More than one billion people live in acute poverty. As a result, they are unable to access proper health care which leads to high levels of infant mortality and lower life expectancy. High levels of infant mortality and low levels of life expectancy will result in a decrease in the growth rate of the populations of poor countries. Therefore, in the future, poverty will slow the growth rate of the world population.

- (a) What is meant by the term 'poverty'? [2 marks]
- (b) Compare Gross National Product (GNP) and Human Development Index (HDI) as appropriate development indices of a country. [8 marks]
- (c) Assess the validity of the statements and the conclusion made in the paragraph above in light of past and current trends in world population growth. [10 marks]

Total 20 marks

6. (a) State TWO features of 'urbanization'. [2 marks]
- (b) Discuss FOUR factors that contribute to urbanization. [12 marks]
- (c) (i) Outline ONE negative environmental impact of urbanization. [2 marks]
- (ii) Suggest TWO solutions to the problem outlined in (c) (i). [4 marks]

Total 20 marks

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MODULE 3

Answer ANY TWO questions.

7. (a) Using an appropriate example, explain what is meant by the term 'natural resource'.
[3 marks]
- (b) Discuss THREE ways in which the natural resource identified in (a) above is important to a **named** Caribbean country.
[9 marks]
- (c) (i) Assess ONE impact of the depletion of the natural resource identified in the country named in (b) above.
[4 marks]
- (ii) Describe a suitable approach that the **named** country may take to manage the natural resource identified.
[4 marks]

Total 20 marks

8. (a) Outline THREE categories of protected areas as classified by the International Union for the Conservation of Nature (IUCN).
[6 marks]
- (b) Explain THREE roles of protected areas in natural resource conservation.
[6 marks]
- (c) Assess the effectiveness of protected areas in conserving natural resources in a **named** Caribbean country.
[8 marks]

Total 20 marks

9. (a) Explain the concept of ecotourism, including in your explanation FOUR characteristics of ecotourism.
[8 marks]
- (b) Discuss FOUR ways in which ecotourism is effective as a tool for natural resource conservation in a **named** Caribbean country.
[12 marks]

Total 20 marks

END OF TEST